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Standpipe recirculation systems for material removal machines

Abstract

Provided are a recirculation system (200) for a material removal machine (102) and a material removal system (100). A material removal system (100) includes a recirculation system (200) in fluid communication with a material removal cabinet (104) housing a material removal machine (102). The recirculation system (200) may have an upper reservoir (300) configured to drain fluid into a lower reservoir (204) before recirculating the fluid. The upper reservoir (300) may have a standpipe (320) configured to allow swarf, debris, and/or other material within the fluid to be entrained prior to flowing over the standpipe (320).

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to standpipe recirculation systems and, more particularly, to standpipe recirculation systems for material removal machines.

BACKGROUND

(2) Some material removal machines (e.g., saws, grinders, polishers, and/or more general material preparation and/or testing machines) use fluid to wash away debris and/or swarf produced during the material removal process. In some cases, a continuous supply of fluid is provided to the material removal machine during the material removal process. Used fluid may be recaptured, recycled, and/or recirculated to provide the continuous supply of fluid for the material removal machine.

(3) Limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art, through comparison of such systems with the present disclosure as set forth in the remainder of the present application with reference to the drawings.

BRIEF SUMMARY

(4) The present disclosure is directed to standpipe recirculation systems for material removal

machines, substantially as illustrated by and/or described in connection with at least one of the figures, and as set forth more completely in the claims.

(5) These and other advantages, aspects and novel features of the present disclosure, as well as details of an illustrated example thereof, will be more fully understood from the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows an example material removal system, in accordance with aspects of this disclosure.

(2) FIGS. 2a and 2b show perspective and front views, respectively, of an example recirculation system of the material removal system of FIG. 1, in accordance with aspects of this disclosure.

(3) FIG. 2c is a perspective view of the recirculation system of FIGS. 2a and 2b with an upper reservoir removed, in accordance with aspects of this disclosure.

(4) FIGS. 3a-3e show various views of the upper reservoir of the recirculation system of FIGS. 2a and 2b, in accordance with aspects of this disclosure.

(5) FIGS. 4 and 5 show examples of alternative standpipes, in accordance with aspects of this disclosure.

(6) FIG. 6 shows an example of a filter used with the upper reservoir of FIGS. 3a-3e, in accordance with aspects of this disclosure.

(7) The figures are not necessarily to scale. Where appropriate, the same or similar reference numerals are used in the figures to refer to similar or identical elements.

DETAILED DESCRIPTION

(8) Some conventional recirculation systems use filters and/or filtration media to separate swarf and/or other particulate material from the fluid that is recirculated to the material removal machine. However, cleaning these filters and/or filtration media (in addition to other components) can be burdensome. Additionally, replacing the filters and/or filtration media can be expensive.

(9) Disclosed example standpipe recirculation systems use a standpipe to separate swarf and/or other particulate material from recirculated fluid, rather than conventional filters and/or filtration media. In some examples, the standpipe blocks fluid from entering a drain in an upper reservoir that leads to a lower reservoir. In some examples, the height of the standpipe may be sufficient to allow swarf, debris, and/or other material within the fluid to be entrained prior to flowing over the standpipe into the lower reservoir.

(10) Some examples of the present disclosure relate to a recirculation system for a material removal machine, comprising: a first reservoir having a floor with a drain; a second reservoir in fluid communication with the first reservoir through the drain; and a standpipe having walls that form a border around the drain within the first reservoir, the walls extending upwards away from the floor to a first height, such that fluid that accumulates on the floor within the first reservoir is blocked from flowing through the drain into the second reservoir until a top of the fluid in the first reservoir exceeds the first height, wherein the first reservoir is configured to permit settlement of material entrained in the fluid prior to the fluid exceeding the first height.

(11) In some examples, the second reservoir is positioned below the drain of the first reservoir to receive the fluid falling through the drain. In some examples, the second reservoir is in fluid communication with a material removal cabinet housing the material removal machine. In some examples, the material removal machine comprises a sectioning saw. In some examples, the system further comprises a recirculation pump configured to pump the fluid from the second reservoir to the material removal cabinet.

(12) In some examples, the system further comprises a removable lid shaped to conform to an

opening in a ceiling of the first reservoir, the removable lid having a lid inlet configured for fluid communication with a cabinet outlet of a material removal cabinet housing the material removal machine, such that fluid can flow from the material removal cabinet to the first reservoir through the cabinet outlet and lid inlet. In some examples, the lid inlet is out of alignment with the drain when the removable lid is fit into the opening in the ceiling of the first reservoir. In some examples, the first reservoir is smaller than the second reservoir. In some examples, the first reservoir and the second reservoir do not include either of a filter or filtration media. In some examples, the height of the standpipe is based on a first capacity of the first reservoir and a second capacity of the second reservoir.

(13) Some examples of the present disclosure relate to a material removal system, comprising: a material removal cabinet housing a material removal machine; a recirculation system in fluid communication with the material removal cabinet, the recirculation system comprising: a first reservoir having a floor with a drain; a second reservoir in fluid communication with the first reservoir through the drain; and a standpipe having walls that form a border around the drain within the first reservoir, the walls extending upwards away from the floor to a first height, such that fluid that accumulates on the floor within the first reservoir is blocked from flowing through the drain into the second reservoir until a top of the fluid in the first reservoir exceeds the first height, wherein the first reservoir is configured to permit settlement of material entrained in the fluid prior to the fluid exceeding the first height.

(14) In some examples, the second reservoir is positioned below the drain of the first reservoir to receive the fluid falling through the drain. In some examples, the second reservoir is in fluid communication with the material removal cabinet housing the material removal machine. In some examples, the material removal system of claim **13**, wherein the material removal machine comprises a sectioning saw. In some examples, the system further comprises a recirculation pump configured to pump the fluid from the second reservoir to the material removal cabinet.

(15) In some examples, the system further comprises a removable lid shaped to conform to an opening in a ceiling of the first reservoir, the removable lid having a lid inlet configured for fluid communication with a cabinet outlet of a material removal cabinet housing the material removal machine, such that fluid can flow from the material removal cabinet to the first reservoir through the cabinet outlet and lid inlet. In some examples, the lid inlet is out of alignment with the drain when the removable lid is fit into the opening in the ceiling of the first reservoir. In some examples, the first reservoir is smaller than the second reservoir. In some examples, the first reservoir and the second reservoir do not include either of a filter or filtration media. In some examples, the height of the standpipe is based on a first capacity of the first reservoir and a second capacity of the second reservoir.

(16) FIG. **1** shows an example of a material removal system **100**. In the example of FIG. **1**, the material removal system **100** includes a material removal machine **102** enclosed in a cabinet **104** and a recirculation system **200** in fluid communication with the cabinet **104**. In the example of FIG. **1**, the material removal machine **102** includes a material removal instrument **108**, such as a (e.g., sectioning) saw blade, abrasive saw, grinder, polisher, and/or some other material removal instrument, for example. In the example of FIG. **1**, the material removal instrument **108** is also encased within a shield **112**. As shown, the shield **112** is attached to a cooling system.

(17) In the example of FIG. **1**, the cooling system includes several coolant hoses **114** attached to the shield **112** through a manifold **116**. Each hose **114** terminates in a nozzle **118**. The nozzles **118** are configured to spray (and/or otherwise provide) coolant into the cabinet **104**. In the example of FIG. **1**, the hoses **114** are configured to receive coolant from a hose inlet **120** that is also attached to the manifold **116**. The hose inlet **120** is in fluid communication with a cabinet inlet **122** through a coolant tube **124**. As shown, the coolant tube **124** is configured to route coolant from the cabinet inlet **122**, through the coolant tube **124** to the hose inlet **120**. The coolant introduced into the cabinet **104** by the coolant system may serve to cool and/or clean the material removal machine

102 and/or other components of the material removal system **100**, while also removing debris, swarf, and/or other particulate material from the cabinet **104**.

(18) In the example of FIG. **1**, the cabinet **104** further includes a cabinet outlet **126**. As shown, the cabinet outlet **126** comprises a porous sieve (and/or mesh, filter, screen, etc.) configured to allow coolant to pass while prohibiting larger particulate matter (e.g., dislodged, freed, and/or unattached components of the material removal machine **102**) from passing. In some examples, the cabinet outlet **126** may omit the sieve, and may simply comprise an opening. In the example of FIG. **1**, the cabinet outlet **126** is in fluid communication with an outlet pipe **128** that leads to an inlet of the coolant recirculation system **200**.

(19) In the example of FIG. **1**, the recirculation system **200** is positioned below the cabinet **104**, such that the force of gravity may be sufficient to propel the coolant through the outlet pipe **128** to the recirculation system **200**. In some examples, the recirculation system **200** may instead be positioned above and/or to the side of the cabinet **104**, and/or some other force (e.g. a pump) may propel the coolant through the outlet pipe **128** from the cabinet **104** to the recirculation system **200**. In the example of FIG. **1** (and FIGS. **2a-2b**), the recirculation system **200** further includes a pump **202** configured to propel recaptured coolant from the recirculation system **200** through a conduit **130** to the cabinet inlet **122** of the cabinet **104**.

(20) FIGS. **2a** and **2b** show perspective and front views, respectively, of an example recirculation system **200**. As shown, the recirculation system **200** includes an upper reservoir **300** and a lower reservoir **204**. The pump **202** is retained by a lower lid **206** of the lower reservoir **204**, and extends some distance into the lower reservoir **204**, so as to intake coolant collected by the lower reservoir **204** and pump the fluid back to the cabinet **104**. For the sake of explanation and/or understanding, the walls of the upper reservoir **300** and lower reservoir **204** have been depicted as transparent in certain drawing figures (e.g., FIGS. **2a-2c** and FIGS. **3b-3c**).

(21) In the examples of FIGS. **2a-2b**, the lower reservoir **204** is significantly larger than the upper reservoir **300** (e.g., three to four times as large). In some examples, the larger size of the lower reservoir **204** may reflect an intention for the lower reservoir **204** to serve as the primary reservoir and/or primary storage for the coolant fluid before recirculation. In some examples, the smaller size of the upper reservoir **300** may make it easier to carry and/or clean, allowing the upper reservoir **300** to serve more as an intermediate and/or filtering reservoir.

(22) In some examples, the lower reservoir **204** (and/or recirculation system **200** in general) may also be sized so that the recirculation system **200** can fit comfortably beneath, and/or approximately within the perimeter of, the material removal cabinet **104**. In FIG. **1**, for example, the cabinet **104** and recirculation system **200** are vertically aligned, with a similar width, such that the material removal cabinet **104** and recirculation system **200** form an approximate rectangle or cube. Such a configuration may allow the material removal system **100** as a whole to be stored together in a neat, compact, and/or aesthetically pleasing arrangement.

(23) In the examples of FIGS. **2a-2b**, the lower reservoir **204** is a hollow cube (and/or cuboid) having a lower floor **208** connected to four lower sidewalls **210** that are themselves connected together to form an upper opening opposite the floor **208**. The lower lid **206** is shaped to fit into that opening, and creates a ceiling for the lower reservoir **204** when inserted into the opening. In some examples, the lower lid **206** may be removable from the lower reservoir **204** (e.g. using lower handles **212**), so that the interior of the lower reservoir **204** may be accessed and/or cleaned. In some examples, the lower lid **206** (and/or lower sidewalls **210**) may include features to make securement and/or removal of the lower lid **206** easier, such as complementary lips, rims, ledges, joints, pillars, recesses, protrusions, and/or flanges. In some examples, fasteners may be used to secure the lower lid **206** to the upper reservoir **300**, and such fasteners may be configured for relatively simple loosening and/or removable in order to allow the lower lid **206** to be removed non-destructively.

(24) In the examples of FIGS. **2a-2b** the upper reservoir **300** is retained on top of the lower

reservoir **204**. However, in some examples, the upper reservoir **300** may be configured for removal from the lower reservoir **204** and/or recirculation system **200**. FIG. 2c, for example, shows a view of the recirculation system **200** with the upper reservoir **300** removed. As shown, a window **214** is visible in the lower lid **206** of the lower reservoir **204** when the upper reservoir **300** is removed. In some examples, the window **214** may be sized and/or configured to receive the upper reservoir **300**. In some examples, coolant fluid may flow from the upper reservoir **300** to the lower reservoir **204** through the window **214** of the lower reservoir **204**.

(25) In some examples, the lower lid **206** may include retaining features to help removably retain the upper reservoir **300** on the lower reservoir **204** (e.g., over and/or in the window **214**). In some examples, the retaining features may enable the upper reservoir **300** to be securely retained on the lower reservoir **204** during operation, and also allow the upper reservoir **300** to be non-destructively removed from the lower reservoir **204** when desired (e.g., for cleaning). In some examples, the upper reservoir **300** may comprise complementary retaining features. Such retaining features may comprise, for example, lips, rims, ledges, joints, pillars, recesses, protrusions, fasteners, and/or flanges.

(26) FIGS. 3a-3e show various views of the upper reservoir **300**. As shown, the upper reservoir **300** is a hollow cube (and/or cuboid) having an upper floor **308** connected to four upper sidewalls **310** that are themselves connected together to form an upper opening opposite the floor **308**. An upper lid **306** is shaped to fit into that upper opening. The upper lid **306** creates a ceiling for the upper reservoir **300** when inserted into the opening, which may help to keep out unwanted debris and/or limit loss of fluid (e.g., via splashing).

(27) In some examples, the upper lid **306** may be removable from the upper reservoir **300** (e.g. using upper handles **312**), so that the interior of the upper reservoir **300** may be accessed and/or cleaned. In some examples, the upper lid **306** may be omitted entirely. In some examples, the upper lid **306** (and/or upper sidewalls **310**) may include features to make securement and/or removal of the upper lid **306** easier, such as complementary lips, rims, ledges, joints, pillars, recesses, protrusions, fasteners, and/or flanges configured to removably secure the upper lid **306** to the upper reservoir **300** such that the upper lid **306** may be removed non-destructively. FIGS. 3d-3e show perspective and top plan views of the upper reservoir **300** with the upper lid **306** removed.

(28) In the examples of FIGS. 3a and 3d-3e, the upper lid **306** includes two lid inlets **302**. In some examples, one of the two lid inlets **302** may be aligned with the outlet pipe **128** to receive coolant fluid from the cabinet **104**. In some examples, having two lid inlets **302** may allow flexibility in positioning the upper reservoir **300** to align with the outlet pipe **128**.

(29) In some examples, the aligned lid inlet(s) **302** may help to position the outlet pipe **128** at an appropriate height, so that there exists an air gap between a top of the fluid retained in upper reservoir **300** and a bottom of the outlet pipe **128**. In some examples, this air gap may help prevent backflow, clogging, and/or other detrimental impact. In some examples, the lid inlet **302** that is not aligned with the outlet pipe **128** may be used as a vent. In some examples, the lid inlet **302** that is not aligned with the outlet pipe **128** may be covered. In some examples, there may be two (or more) outlet pipes **128**, with each outlet pipe **128** aligned with a lid inlet **302**. In some examples, the upper lid **306** may include only one lid inlet **302** or three or more lid inlets **302**.

(30) In the examples of FIGS. 3b-3e, the upper reservoir **300** includes a drain **304** in its upper floor **308**. With the upper reservoir **300** positioned atop the window **214** in the lower lid **206**, coolant fluid flowing through the drain **304** will flow into the lower reservoir **204** under the force of gravity. In some examples, a drain pipe may help guide fluid flowing through the drain **304** into the lower reservoir **204**.

(31) In the examples of FIGS. 3b-3e, the upper reservoir **300** also includes a standpipe **320**. While one standpipe **320** is shown, in some examples, multiple standpipes **320** (and/or multiple drains **304**) may be within the upper reservoir **300**. In the examples of FIGS. 3d-3e, the standpipe **320** and drain **304** are offset and/or out of alignment with the lid inlets **302**, so that fluid entering the upper

reservoir **300** through the lid inlets **302** will not flow directly into the drain **304**. In some examples, fluid (and/or the outlet pipe **128**) may be directed through the lid inlets **302** to a position within the upper reservoir **300** opposite the standpipe **320** and/or drain **304**, so as to allow fluid time and/or space to settle and/or become less turbulent before encountering the standpipe **320** and/or drain **304**.

(32) As best shown in FIGS. **3b** and **3c**, the standpipe **320** has a base **324** underneath the upper floor **308**. In some examples, the base **324** may be affixed to an underside of the upper floor **308** of the upper reservoir **300**. In some examples, the base **324** (and/or standpipe **320**) may be affixed to a top side of the upper floor **308** of the upper reservoir **300**. In some examples, the standpipe **320** may be molded with the upper floor **308**. In some examples, the standpipe **320** may be friction fit into the drain **304**.

(33) In the examples of FIGS. **3b-3c**, the standpipe **320** has walls **322** extending upwards from the base **324**, through the drain **304**, and into the upper reservoir **300**. Within the upper reservoir **300**, the walls **322** extend upwards away from the upper floor **308**. As shown, the walls **322** form a border around the drain **304** to block fluid on the upper floor **308** from flowing directly into the drain **304**.

(34) In the examples of FIGS. **3b-3c**, the walls **322** of the standpipe **320** extend upwards away from the floor **308** to a standpipe height. In some examples, only coolant fluid that accumulates and/or pools in the upper reservoir to a height greater than the standpipe height can flow over the walls **322** of the standpipe **320** and through the drain **304** (and window **214**) into the lower reservoir **204**. In some examples, the time required for coolant entering the upper reservoir **300** to reach the standpipe height may be enough to allow swarf, debris, and/or other particulate material entrained in the coolant fluid to settle to the upper floor **308** of the upper reservoir **300**. As a result, the swarf, debris, and/or other particulate material settles below the top of the standpipe **320**, and is not carried through the drain **304** to the lower reservoir **204**. Thus, in some examples, the standpipe **320** may enable the recirculation system **200** to separate swarf, debris, and/or other particulate matter from the recirculating coolant fluid without the use of conventional filters and/or filtration media.

(35) In the examples of FIGS. **3b-3c**, the standpipe height is between approximately a quarter and a third of the height of the upper reservoir **300**. In some examples, the standpipe height may be more or less than this height. In some examples, the standpipe height may be larger or smaller. In some examples, the standpipe height is less than an upper height threshold at which there is a risk of starving the pump **202** of coolant fluid. For example, at an upper height threshold, the standpipe **320** may cause the upper reservoir **300** to retain so much of the fluid that there is a risk the fluid level in the lower reservoir **204** will remain too low for the pump **202** to intake fluid for recirculation to the cabinet **104** (i.e., thereby starving the pump **202**). In some examples, the upper height threshold is dependent on the relative sizes of the upper reservoir **300** and lower reservoir **204**, as well as how far the pump **202** extends into the lower reservoir **204**, a total fluid amount in the system, and/or other factors.

(36) FIG. **4** shows an example of another example standpipe **420** that may be used instead of the standpipe **320** illustrated in FIGS. **3a-3e**. As shown, the example standpipe **420** has a base **424** affixed to the upper floor **308** of the upper reservoir **300** with fasteners **426** that extend through compressible springs **428**. The springs **428** serve to bias the standpipe **420** upwards. The fasteners **426** are mounted with shafts that extend downwards from the upper floor **308**, such that the base **424** (and connected standpipe **420**) can move downwards on and/or over the fasteners **426** when supplied with sufficient downward force (e.g., via an operator) to overcome the bias of the springs **428**. Thus, the standpipe **420** may be moved to a lower height via the fasteners **426** and springs **428**.

(37) In some examples, the fasteners **426** may have shaft of sufficient length to allow the standpipe **420** to be moved down to half (or less) the standpipe height, a quarter (or less) the standpipe height,

the height of the upper floor **308** and/or drain **304**, or slightly above the height of the upper floor **308** and/or drain **304**. Once the standpipe height is lowered, the upper reservoir **300** may be more easily evacuated of fluid. Once the fluid is evacuated, the upper reservoir **300** may be more easily removed from the recirculation system **200** (e.g., for cleaning). After the fluid is gone, and the downward force is removed, the biasing force of the springs **428** may push the standpipe **420** back up to the original standpipe height.

(38) FIG. 5 shows another example standpipe **520**. As shown, the standpipe **520** is configured to be pulled up rather than pushed down to remove any remaining fluid in the upper reservoir **300**. As shown, the standpipe **520** has a lower portion **528** that is normally positioned below the base **524**, and underneath the upper floor **308** of the upper reservoir **300**. When pulled up (e.g., by an operator), a lower portion **528** of the alternative standpipe **520** will enter the upper reservoir **300** from below, and form the border around the drain **304**.

(39) In the example of FIG. 5, the lower portion **528** of the second alternative standpipe **520** has slits **526**. In some examples, the slits **526** may be made narrow enough to operate as a coarse filter. In some examples, the slits **526** may instead be holes, or other shaped openings. In some examples, when the standpipe **520** is raised, the fluid in the upper reservoir **300** can flow through the slits **526** and into the drain **304**, regardless of the fluid height. As shown, the lower portion **528** of the walls **522** additionally has a jutting protrusion **530** that prevents the second alternative standpipe **520** from being pulled up past the base **524**. Thus, in some examples, the second alternative standpipe **520** may allow an operator to easily drain fluid from the upper reservoir **300** by raising the lower portion **528** of the second alternative standpipe **520**, and allowing the fluid to drain through the slits **526**.

(40) The example recirculation systems **200** disclosed herein enable separation of swarf, debris, and/or other particulate material from recirculated fluid without the use of conventional filters and/or filtration media. The lack of conventional filters and/or filtration media may facilitate easier cleaning and/or lower cost. However, conventional filters and/or filtration may also be used with the example recirculation systems **200** described herein, if desired.

(41) FIG. 6, for example, shows an example of a filter **600** positioned within the upper reservoir **300**. As shown, the filter **600** extends horizontally across the upper reservoir, from sidewall **310** to sidewall **310**, at a height above the upper floor **308**, and below the height of the standpipe **320**. In some examples, the filter **600** may be tightly fitted around the standpipe **320**. In some examples, the outlet pipe **128** of FIG. 1 drains the fluids from the cabinet **104** to the volume between the upper floor **308** and the filter **600**. Thus, fluid rising from the upper floor **308** to the standpipe height may have to pass through the filter **600** before going through the drain **304**. In some examples, the filter **600** may have one or more openings aligned with the inlet(s) **302**, to allow fluid to enter the upper reservoir **300** without passing through the filter **600**. In some examples, the outlet pipe **128** may extend into the upper reservoir **300** (e.g., through an inlet **302**) to better direct fluid to the opening(s) in the filter **600**.

(42) While the present method and/or system has been described with reference to certain implementations, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the present method and/or system. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. Therefore, it is intended that the present method and/or system not be limited to the particular implementations disclosed, but that the present method and/or system will include all implementations falling within the scope of the appended claims.

(43) As used herein, “and/or” means any one or more of the items in the list joined by “and/or”. As an example, “x and/or y” means any element of the three-element set {(x), (y), (x, y)}. In other words, “x and/or y” means “one or both of x and y”. As another example, “x, y, and/or z” means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. In other words, “x,

y and/or z” means “one or more of x, y and z”.

(44) As utilized herein, the terms “e.g.,” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations.

(45) As used herein, the terms “about” and/or “approximately,” when used to modify or describe a value (or range of values), position, orientation, and/or action, mean reasonably close to that value, range of values, position, orientation, and/or action. Thus, the examples described herein are not limited to only the recited values, ranges of values, positions, orientations, and/or actions but rather should include reasonably workable deviations.

(46) As used herein, the terms “coupled,” “coupled to,” and “coupled with,” each mean a structural and/or electrical connection, whether attached, affixed, connected, joined, fastened, linked, and/or otherwise secured. As used herein, the term “attach” means to affix, couple, connect, join, fasten, link, and/or otherwise secure. As used herein, the term “connect” means to attach, affix, couple, join, fasten, link, and/or otherwise secure.

(47) As used herein, the term “fluid,” when used as a noun, refers to a free-flowing deformable substance with no fixed shape, including, inter alia, liquid (e.g., water, solution, etc), and/or plasma.

Claims

1. A recirculation system for a material removal machine, comprising: a first reservoir having an opening and a floor with a drain; a removable lid shaped to conform to the opening in the first reservoir, the removable lid having a lid inlet, such that fluid can flow into the first reservoir through the lid inlet from a cabinet outlet of a material removal cabinet housing the material removal machine; a second reservoir in fluid communication with the first reservoir through the drain; and a standpipe having walls that form a border around the drain within the first reservoir, the walls extending upwards away from the floor to a first height, such that fluid that accumulates on the floor within the first reservoir is blocked from flowing through the drain into the second reservoir until a top of the fluid in the first reservoir exceeds the first height.
2. The recirculation system of claim 1, wherein the second reservoir is positioned below the drain of the first reservoir to receive the fluid falling through the drain.
3. The recirculation system of claim 1, wherein the second reservoir is in fluid communication with the material removal cabinet housing the material removal machine.
4. The recirculation system of claim 3, wherein the material removal machine comprises a sectioning saw.
5. The recirculation system of claim 3, further comprising a recirculation pump configured to pump the fluid from the second reservoir to the material removal cabinet.
6. The recirculation system of claim 1, wherein the opening is a first opening, the recirculation system further comprising a second lid shaped to conform to a second opening in the second reservoir, the second lid having a window sized to receive the first reservoir.
7. The recirculation system of claim 6, further comprising a recirculation pump retained by the second lid.
8. The recirculation system of claim 1, wherein the lid inlet is out of alignment with the drain when the removable lid is fit into the opening of the first reservoir.
9. The recirculation system of claim 1, wherein the first reservoir is smaller than the second reservoir.
10. The recirculation system of claim 1, wherein the first reservoir and the second reservoir do not include either of a filter or filtration media.
11. The material removal system of claim 1, wherein the lid inlet is out of alignment with the drain when the removable lid is fit into the opening of the first reservoir.
12. A material removal system, comprising: a material removal cabinet housing a material removal

machine; and a recirculation system in fluid communication with the material removal cabinet, the recirculation system comprising: a first reservoir having a lid an opening and a floor with a drain, a removable lid shaped to conform to the opening in the first reservoir, the removable lid having a lid inlet, such that fluid can flow into the first reservoir through the lid inlet from a cabinet outlet of the material removal cabinet housing the material removal machine, a second reservoir in fluid communication with the first reservoir through the drain, and a standpipe having walls that form a border around the drain within the first reservoir, the walls extending upwards away from the floor to a first height, such that fluid that accumulates on the floor within the first reservoir is blocked from flowing through the drain into the second reservoir until a top of the fluid in the first reservoir exceeds the first height.

13. The material removal system of claim 12, wherein the second reservoir is positioned below the drain of the first reservoir to receive the fluid falling through the drain.

14. The material removal system of claim 12, wherein the second reservoir is in fluid communication with the material removal cabinet housing the material removal machine.

15. The material removal system of claim 14, wherein the material removal machine comprises a sectioning saw.

16. The material removal system of claim 14, further comprising a recirculation pump configured to pump the fluid from the second reservoir to the material removal cabinet.

17. The material removal system of claim 12, wherein the opening is a first opening, the recirculation system further comprising a second lid shaped to conform to a second opening in the second reservoir, the second lid having a window sized to receive the first reservoir.

18. The material removal system of claim 17, further comprising a recirculation pump retained by the second lid.

19. The material removal system of claim 12, wherein the first reservoir is smaller than the second reservoir.

20. The material removal system of claim 12, wherein the first reservoir and the second reservoir do not include either of a filter or filtration media.
